

This function gets called by RunPython when migrating forward.

The `apps.get_model()` method returns a `Model` class, whose state is the same as the current migration.

```
def de_westernize_names(apps, schema_editor):
    # Can't import Promoter Model, the current version may be newer than
    # the one in this migration. Use historical version instead.
    Promoter = apps.get_model("promoters", "Promoter")
    for promoter in Promoter.objects.all():
        promoter.full_name = f"{promoter.first_name} {promoter.last_name}"
        promoter.common_name = promoter.first_name
        promoter.save()
```

This function gets called by RunPython when migrating backward. It is similar to `de_westernize()`, except it undoes the data translation.

Data translation for Promoter requires enumerating all objects, and for each one, constructing the `.full_name` field based on the `.first_name` and `.last_name` fields and then copying the `.first_name` field into the `.common_name` field.

```
def re_westernize_names(apps, schema_editor):
    Promoter = apps.get_model("promoters", "Promoter")
    for promoter in Promoter.objects.all():
        promoter.first_name = promoter.common_name
        length = len(promoter.first_name)
        promoter.last_name = promoter.full_name[length + 1:]

        promoter.save()

class Migration(migrations.Migration):
    dependencies = [
        ("promoters", "0003_promoter_common_name_promoter_full_name"),
    ]

    operations = [
        migrations.RunPython(
            de_westernize_names,
            re_westernize_names
        ),
    ]
```

RunPython is an operation that takes two function references: the first gets called when migrating forward, and the second when migrating backward.

Your data translation migration script is ready. The full process, including removal of the first and last name fields, is three steps, but that doesn't mean you should run them all together. Now is a good time to test. If you've made an error in your script, you might lose data. Your data is four sample lines of Promoters and can easily be re-entered. The real world is seldom that forgiving. You should take a backup of your database and test your scripts by running the migrate management command. Once you've done that, validate that the script did what you wanted. If it didn't, restore your backup, fix your bugs, and try again. Once you're satisfied that 0004 is working, you can move to the third, and final, step in the process.

The last step is to remove the first and last name fields. This step shouldn't contain any surprises. Open the model file, delete the appropriate lines, run `makemigrations`, and then migrate. The final state of `Promoter` is shown in the following listing.